Shashank Kumar

Full-Stack & Applied AI Engineer | AI Safety | LLM Systems
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Summary

Engineer with five years shipping production software, now working across applied cryptography and LLM systems. Author of CHRONOS, a Rust implementation of an ephemeral agent combining homomorphic encryption, a verifiable delay function, and a Groth16 erasure proof (8,267 R1CS constraints, 128-byte proofs verifying in 1 ms). Delivered a pre-launch security audit that surfaced a PCI-DSS violation and an authentication bypass. Rated top 5% on Alignerr's adversarial evaluation programme; Top Rated on Upwork. Owns problems end to end: design, Rust and Python implementation, benchmarking, and precise reporting of what is and is not proven.

Technical Skills

Languages: Rust, Python, TypeScript, JavaScript, C++, C#, SQL, Bash

Cryptography: Groth16 zk-SNARKs (arkworks), Poseidon hash, verifiable delay functions, BLS12-381, ML-DSA, AES-256-GCM, authenticated encryption

AI / ML: Multi-agent orchestration (FSM/DAG), RAG pipelines, prompt engineering, LLM fine-tuning (LoRA, DPO), RLHF, adversarial red-teaming, FHE inference (TFHE-rs), formal verification (Z3 SMT), YOLOv9, PPO/RL, NVIDIA Isaac Sim

Backend: FastAPI, Node.js, Express, GraphQL, gRPC, PostgreSQL, pgvector, TimescaleDB, Redis, MongoDB, REST APIs

Frontend: Next.js, React, Tailwind CSS, HTML5, CSS3

Infrastructure: Docker, GitLab CI/CD, GitHub Actions, Ansible, AWS, Azure, Prometheus, Grafana, Hugging Face, OpenAI and Anthropic APIs

Security: PCI-DSS auditing, threat modelling, mTLS, JWT, HMAC-SHA256, secrets management

Experience

Community Member - Cohere (Remote) Sep 2026 – Present

- Admitted to the Cohere Open Science Community based on a technical review of open-source verifiable AI projects.

Full Stack AI Engineer - BitNorm (Netherlands, Remote) Nov 2025 – Present

- Security-audited a pre-launch Rust and GraphQL platform; identified critical vulnerabilities in payment handling, authentication, and secrets management, including a PCI-DSS compliance violation.
- Delivered a prioritised remediation plan accepted by the engineering team: payment tokenization, GraphQL field masking, secrets management, and CORS hardening.
- Diagnosed database query inefficiencies in nested GraphQL resolvers and specified a batching pattern to reduce per-request round-trips.

AI Systems Evaluator and Red Teamer - Alignerr (Remote) Apr 2025 – Present

- Rated **top 5%** (platform-issued badge) across 1,000+ adversarial evaluation tasks.
- Authored 500+ adversarial red-team prompts against pre-release frontier models, targeting jailbreak resistance, factual grounding, and instruction-following.
- Evaluated three or more pre-release LLMs for factual accuracy and logical coherence ahead of production release.

Freelance AI and Full-Stack Developer - Upwork (Remote) May 2024 – Present

- Delivered LLM safety evaluation and adversarial testing for a client platform; **client-measured 25% improvement** in their internal safety scoring.
- Built a healthcare support chatbot on GPT-4 and Flask covering 500+ patient FAQs; 92% response accuracy and an 80% reduction in response time.
- Built custom RAG pipelines improving answer accuracy on client document sets; maintained **Top Rated** status on the platform.
- Annotation and model-evaluation work via Outlier AI (2024–present) alongside development engagements.

Selected Projects

CHRONOS - Ephemeral AI Agent with Verifiable Self-Termination Rust, arkworks, TFHE-rs

- Composed homomorphic encryption, a Wesolowski verifiable delay function, and a Groth16 erasure proof so an agent's key is released only by sequential work and its destruction is checkable by a third party. Hand-built Poseidon-128 over BN254 and a Poseidon-based AEAD for in-circuit authenticated decryption.
- Measured: **8,267 R1CS constraints** (all load-bearing), **128-byte proofs verifying in 1 ms**, VDF throughput constant in T at $\sim$**505k squarings/sec**; containment invariants checked exhaustively over 1,728 abstract states at startup.
- Documented limitations: single-party trusted setup, and FHE inference at small scale only (2.29 s per homomorphic multiplication). github.com/ChronosResearch/project-chronos

PAAC - Provably Aligned AI Core Python, Z3 SMT

- Deterministic safety wrapper using the Z3 SMT solver to verify constraints on self-modifying agent code before execution rather than after. **382 passing tests** across the constraint layer, solver integration, and failsafe paths. github.com/ChronosResearch/paac

CellHawk - GPS-Denied Navigation Research Stack Python, C++, EKF

- Position estimation from ambient cellular signal strength via RSSI multilateration fused through an Extended Kalman Filter, requiring no additional hardware; swarm coordination layer and 3D voxel path planner with A* replanning.
- **Simulation only:** validated against JSBSim flight dynamics and a synthetic cellular dataset. No hardware flight testing performed.

PyRobotics - Robotics Algorithm Reference Library Python, NumPy

- Ten-plus module library implementing RRT*, A*, LQR, MPC, MPPI, EKF, UKF, particle filter, pose-graph SLAM, and a policy-gradient agent with a pure-NumPy neural network. Simulation only; tests cover simulation, perception, and control. github.com/sidthebuilder/PyRobotics

Epistemic Infrastructure and **Solar Phantom** Python, Z3, AeroSandbox

- **Epistemic Infrastructure:** structured reasoning layer addressing retrieval-augmented generation failure modes that ranking improvements alone do not fix, with replayable traces.
- **Solar Phantom:** multidisciplinary design optimisation solving for solar-UAV geometry that closes a 24-hour energy balance; latitude feasibility maps and day-of-year survival windows. Physics via AeroSandbox; not yet validated against a reference aircraft.

Preprints

None of the following are peer-reviewed. Full list: Google Scholar.

- *Project CHRONOS: A Fully Homomorphic Ephemeral AI Agent* - SSRN 6950898, Jun 2026.
- *The Provably Aligned AI Core: A Deterministic Safety Wrapper* - SSRN 6879218, Jun 2026.
- *The Zero-Training Enterprise Brain: A Live Semantic Reasoning Architecture* - SSRN 6842778, May 2026.
- *Building Epistemic Infrastructure for Enterprise AI Search* - SSRN 6585978, Apr 2026.
- *Beyond Single Prompts: Engineering a Multi-Agent Workgroup for End-to-End Tabular Data Science* - technical report, Mar 2026.

Education

B.Sc. Computer Science - University of the People Apr 2025 – Expected 2027
In progress. Coursework in algorithms, data structures, and software engineering.

B.A. (English, History, Hindi) - Swami Vivekanand Subharti University, Meerut 2021 – 2025

Additional

Certifications: CS50 Introduction to Computer Science (Harvard, edX, 2024) • Google Analytics and Digital Marketing (Google Digital Garage, 2023) • Scale AI Annotation Specialist

Languages: English (fluent), Hindi (native)

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